



CampusHub

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1. Introduction

1.1 System Purpose

A college management system is designed to streamline and centralize the day-to-day operations of a college, making administrative tasks faster, more accurate, and easier to manage. It helps handle student admissions, attendance, examinations, fee payments, and academic records in one organized platform. By connecting students, teachers, and administrators, the system improves communication, reduces paperwork, and ensures that important information is accessible when needed. Overall, it enhances efficiency and supports better decision-making across the institution.

1.2 System Scope

The scope of a college management system covers almost every functional area of a college's operations. It includes managing student admissions, enrollment, attendance, examinations, grading, and academic records. It also supports administrative functions like staff management, payroll, timetable scheduling, fee collection, inventory, and library services. Beyond administration, the system can enable communication between students, faculty, and management, provide portals for online learning and assignments, and generate reports for planning and decision-making. As colleges grow, the system can expand to support multiple campuses, integrate with other digital tools, and provide analytics to improve overall institutional performance.

1.3 System Overview

1.3.1 System Context

The system has been designed with multiple modules to efficiently manage academic and administrative activities within a college environment. The primary modules of the system include the Admin Module, Faculty Module, and Student Portal Module. Together, these modules form the overall structure of the College Management System.

The Admin Module serves as the core administrative component of the system and is responsible for managing institutional operations. This includes handling student admissions, faculty records, course and subject management, department details, timetable scheduling, examination management, fee processing, and overall system maintenance. The admin has full control over data management and user access rights to ensure smooth functioning of the institution.

The Faculty Module is designed to support teaching staff in managing academic responsibilities. Through this module, faculty members can view assigned courses, upload study materials, manage attendance, enter internal assessment marks, and communicate important academic information to students.

The Student Portal Module provides an interactive web-based interface for students. Through this portal, students can access personal academic details, view course registrations, attendance records, examination schedules, results, fee payment status, and notifications issued by the administration or faculty.

The system enables seamless interaction among administrators, faculty members, and students, ensuring efficient information flow and reducing manual effort. By automating core academic and administrative processes, the College Management System enhances transparency, accuracy, and operational efficiency within the institution.

1.3.2 System Function

- The system will be able to manage student admissions and maintain student records.
- The system will be able to manage faculty details and course information.
- The system will be able to record and track student attendance, examination details, and academic performance.
- The system will be able to generate reports for students, faculty, courses, attendance, results, and fee details.

1.3.3 User Characteristics

The users of the system will include students, faculty members, and administrative staff. Students may consist of newly admitted students, currently enrolled students, and alumni with limited system access. Faculty members will include teaching staff responsible for academic activities, while administrative users will include office staff and system administrators responsible for managing institutional operations.

The administrative users will primarily use desktop or laptop computers running the Windows operating system to perform administrative tasks on the system. Faculty members may access the system using desktops or laptops for academic management purposes. The system must also accommodate a variety of devices used by students, including desktops, laptops, tablets, and mobile devices running different operating systems, to ensure smooth access to the web portal.

The system should be user-friendly and accessible to users with varying levels of technical knowledge, ensuring that all users can effectively interact with the system to perform their respective tasks.

1.4 Definitions

College Management System:

- College Registrar
- Employee
- Student
- Web Portal for Users

Actors:

- College Registrar
- Employee
- Student

1.5 Technologies To Be Used

- React + CSS (for styling)
- Spring Boot
- PostgreSQL

2. Requirements

2.1 External Interfaces

2.1.1 Hardware Interfaces

RAM: Minimum 8 GB

Network: Stable LAN / Internet connectivity

Processor: Intel Core i3 or higher

Storage: Minimum 100 GB free disk space

Display: 1024 × 768 resolution or higher

Input Devices: Keyboard and Mouse

2.1.2 Software Interfaces

Operating System: Windows / Linux / macOS / Android / iOS

Web Browser: Google Chrome, Mozilla Firefox, Microsoft Edge

Development Tools

IDE: Eclipse, VS Code

Backend Technology: Java (Spring Boot)

Database Tools: PostgreSQL

Version Control: Git

Frontend Technologies: React

2.1.3 User Interfaces

First of all, the user has to log in to the system using valid credentials to access the portal.

Secondly, students can view and manage their academic information such as courses, attendance, examination schedules, and results.

And third, faculty members can update attendance, upload study materials, and enter internal assessment marks, while administrative users can manage student, faculty, and course details through the system interface.

2.2 Functions

2.2.1 Functional Requirements

System:

The system should have a login facility for all users.

A login interface should appear when the system is invoked.

The system should maintain records of students, faculty members, courses, departments, and academic activities.

The system should keep track of academic details such as attendance, examinations, results, and fee information.

Admin:

The Admin should have all types of authority over the system.

The Admin should manage student admissions and student records.

The Admin should maintain faculty details, course information, and department data.
The Admin should manage timetables, examination schedules, and academic calendars.
The Admin should generate reports related to students, faculty, attendance, results, and fees.

Faculty:

Faculty log in through the portal.
Faculty views assigned subjects and class details.
Faculty records and updates student attendance.
Faculty upload study materials and assignments.
Faculty enters internal assessment marks and examination results.

Student:

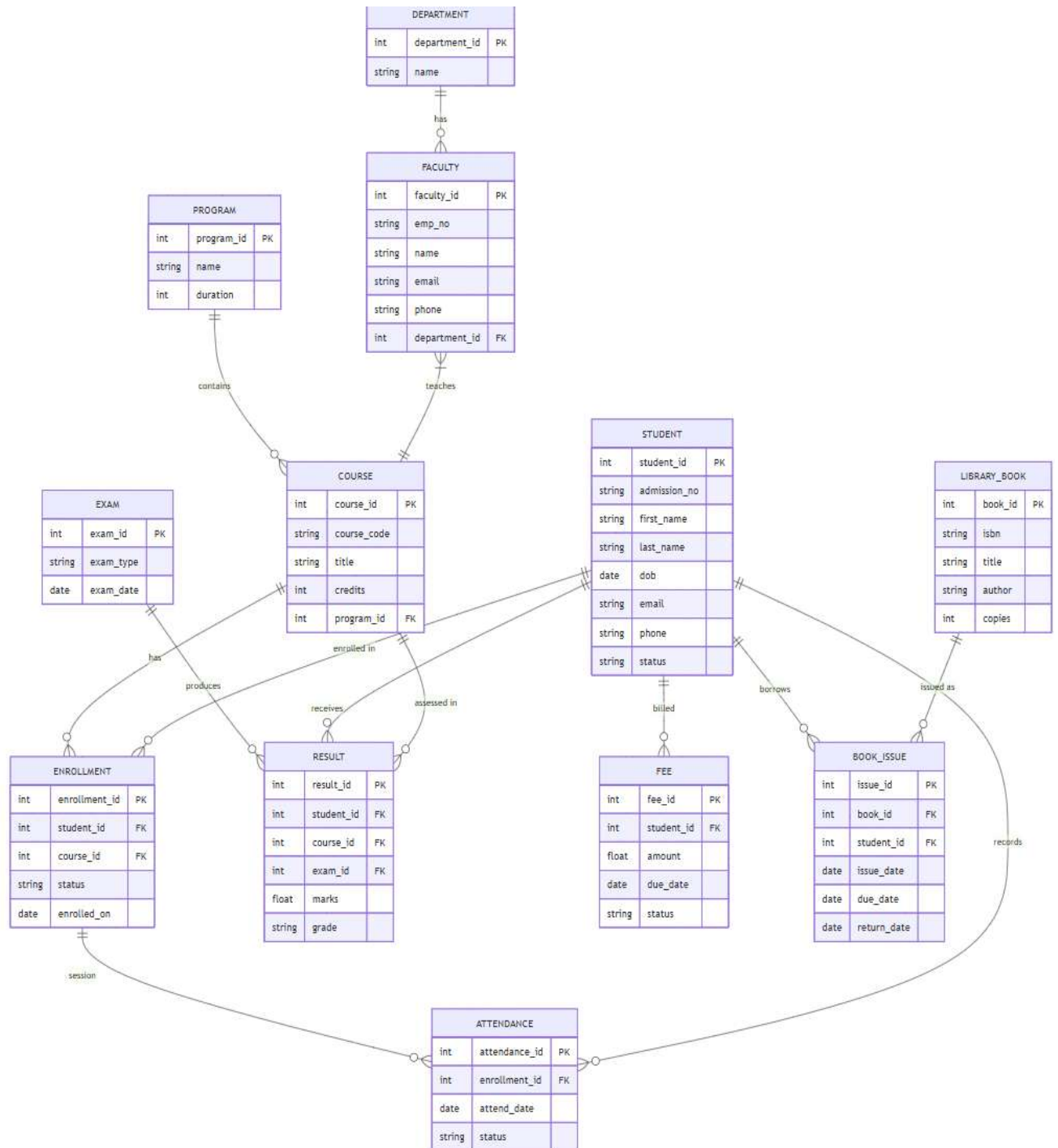
Students log in through the portal.
Students view personal profile and academic details.
Students view course information, attendance records, and examination schedules.
Student views results and notifications published by the administration or faculty.

2.3 Design Constraints

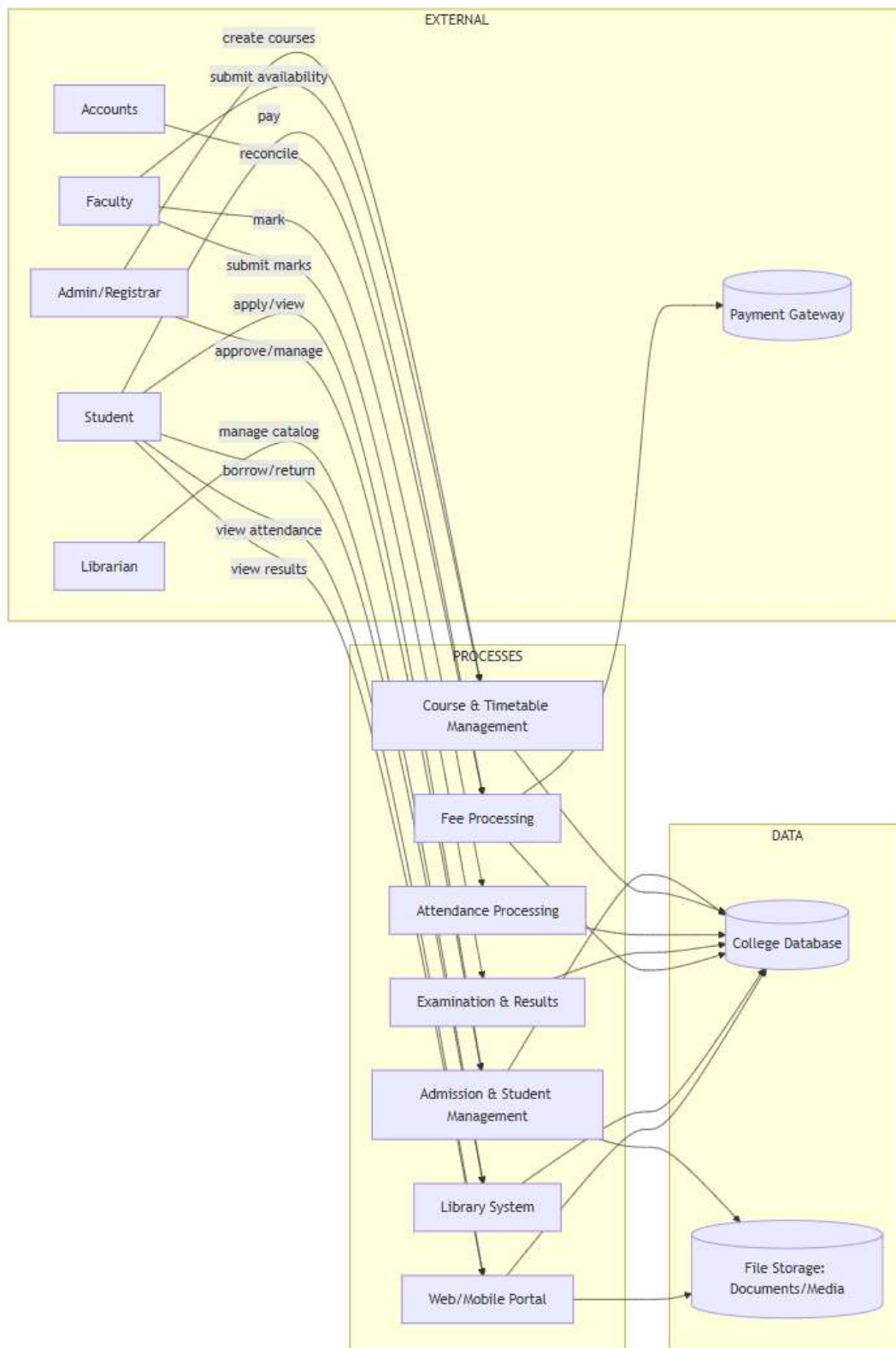
The system will be constrained by operating software of the host system and will need to be able to function on the different internet servers:

1. The system will need to function on major internet operating software including Internet explorer, Firefox, Chrome, Safari, Opera, and Android.
2. The system will be constrained operating software of the host computer which is windows.

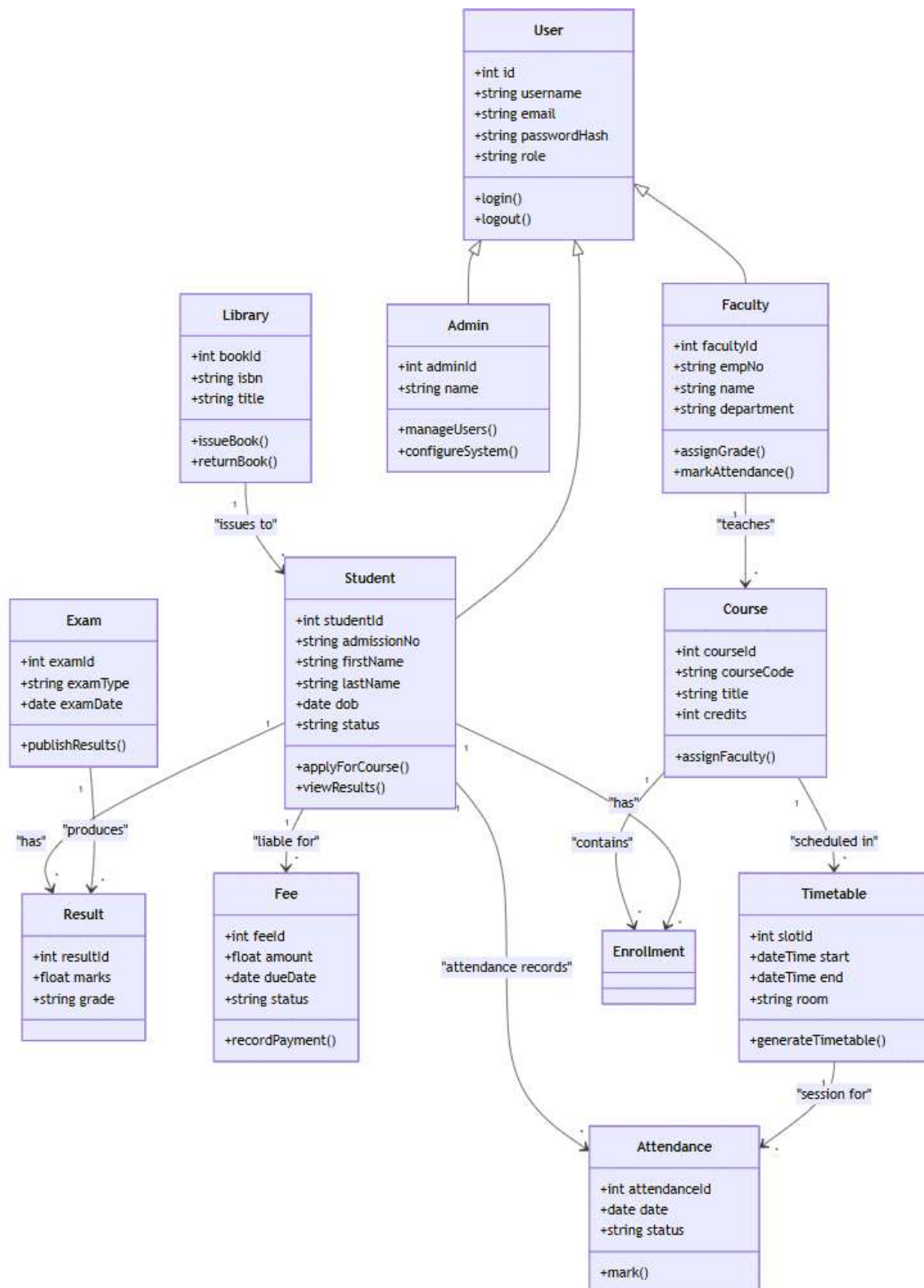
2.4 Entity Relationship Diagram



2.5 Data Flow Diagram (Level 1)



2.6 Class Diagram



3. Conclusion

CampusHub, the College Management System, has been designed and developed to provide an efficient, reliable, and user-friendly platform for managing academic and administrative activities within an educational institution. The system integrates essential modules such as administration, faculty management, and student services into a single centralized solution, thereby reducing manual effort and improving overall operational efficiency.

By automating processes such as student admissions, course management, attendance tracking, examination handling, and result generation, CampusHub ensures accuracy, transparency, and timely access to information for all users. The system facilitates smooth interaction between administrators, faculty members, and students, enabling better communication and streamlined workflows.

CampusHub also offers scalability and flexibility, allowing institutions to adapt the system to their specific requirements and future growth. Overall, the implementation of CampusHub contributes to enhanced productivity, better data management, and improved decision-making, making it a valuable solution for modern educational institutions.